Read motion from the slope

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Velocity / ES: velocidad / PT: velocidade

How fast and which way. The rate position changes.

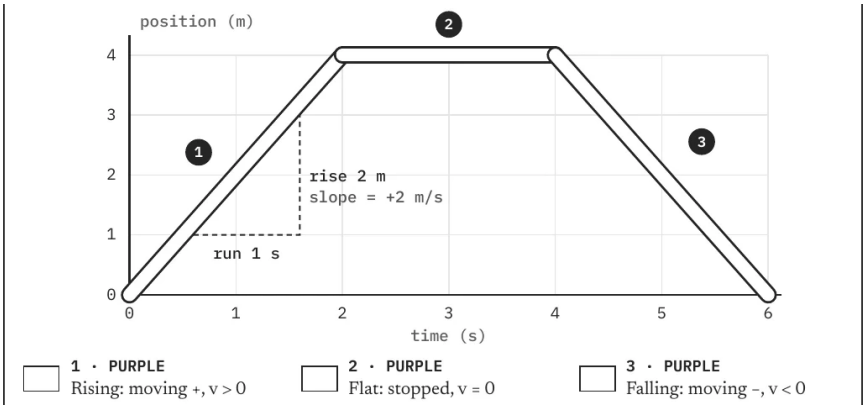
Also on your note page: Speed, Slope, Average speed.

B READ THE RELATIONSHIP

Velocity tells how fast position changes and in which direction.

Average velocity = displacement ÷ elapsed time

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart goes from 8 m to 2 m in 3 s. Right is positive. Find its velocity.

Given $x_i = 8$ m $x_f = 2$ m $\Delta t = 3$ s

Rule $v = \Delta x \div \Delta t$

Work $\Delta x = 2 - 8 = -6$ m $v = -6$ m ÷ 3 s

Answer −2 m/s 2 m/s to the left

E SAVE FOR THE FINAL MEETING

Two position readings and the time between them give XJ's velocity: the slope of its position–time graph. That tells us how fast it closes in, and from which side.

Read the physics: Velocity

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Find the start and finish on a position–time graph. Read position vertically and time horizontally. A rising line means position increases. A flat line means the object stays in one place.

02 HOW IT WORKS

Divide change in position by elapsed time. The slope is average velocity over that interval. A falling line has negative slope: the object moves toward smaller positions.

03 WHAT TO CHECK

Speed uses total distance. A return trip can have zero average velocity even when the object moved.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart goes from 8 m to 2 m in 3 s. Right is positive. Find its velocity. –2 m/s 2 m/s to the left.

05 CONNECT TO XJ

Two position readings and the time between them give XJ's velocity: the slope of its position–time graph. That tells us how fast it closes in, and from which side.

SAY IT BEFORE YOU WRITE

What does the sign of the slope tell you? The line slopes ___. The object moves ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Velocity / ES: velocidad / PT: velocidade

Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, velocity means …	What does the minus sign in −2 m/s tell you about the cart? The sign tells me the cart is moving ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Sketch a falling position–time line. Label the signed rise, positive run and velocity unit.

Labels you can use: rise · run · slope.



Tell a partner: The line slopes ___. The object moves ___.

On assignment: Velocity

SECTION E / BACK / FINAL MEETING ONLY Name _____

Radar sees XJ at +12 km, then +6 km three seconds later. Positive points away from Earth.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Radar operator**

Send a signed velocity to the tracking console. Mark the two fixes on a sloping line so the next operator can check the sign.

☐ **Traffic broadcaster**

Give viewers a speed and direction. Your producer points at the two readings and asks whether they prove XJ is slowing.

Word bank: rise · run · slope.

Try: “Our desk recommends ___. This mark shows ___.”

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

“We first said ___. Now we ___ because ___.”